

Municipal Finance Control Layer

Modernizing municipal finance without replacing the systems already in place.

An AI-assisted control and integration architecture connecting accounting, utility, banking, payroll, grant, project and governing evidence.

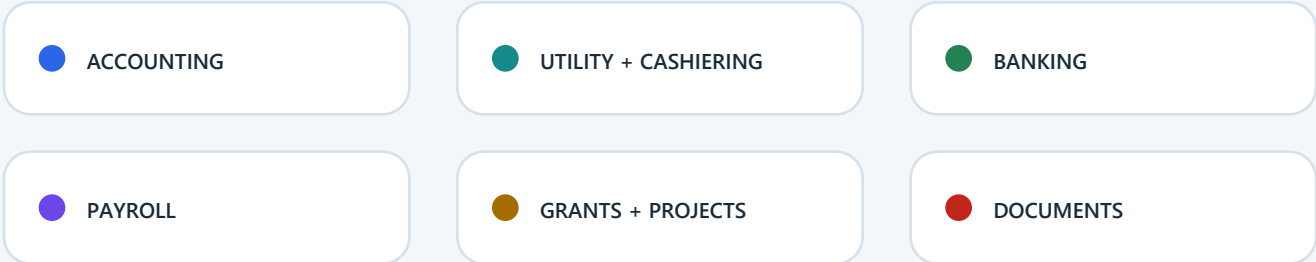
PINEHAVEN SYNTHETIC CASE STUDY

EXISTING AND LEGACY SOFTWARE

PURPOSE

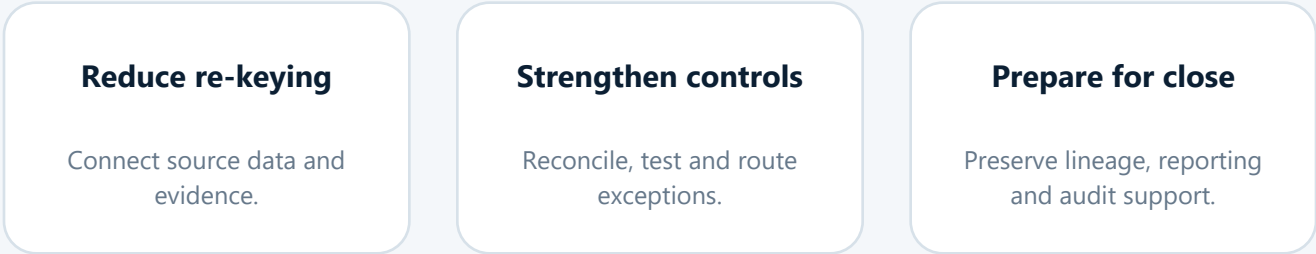
One controlled layer across the municipal finance lifecycle

MFCL integrates data from existing systems while those systems remain the authoritative systems of record.



Designed to integrate with existing and legacy QuickBooks Desktop environments; functionally tested with QuickBooks Desktop 2022 in a controlled test environment.

No replacement ERP or accounting-system purchase is required to use the architecture concept.



SYSTEMS ARCHITECTURE

Systems flow: source evidence to reporting



PUBLIC TECHNICAL ARCHITECTURE

A practical technical foundation for controlled municipal integration

CONNECT + INGEST

- Version-aware adapters
- Incremental synchronization
- Source IDs + provenance
- Watermarks / cursors
- Duplicate + out-of-order controls
- Idempotent processing
- Void / change / delete preservation
- Secure-file fallback

STRUCTURED DATA

- Relational / SQL-compatible records
- JSON-compatible normalized events
- Evidence / document registry
- Effective-dated authority records
- Versioned mappings and rules

HIGH-LEVEL CANONICAL DIMENSIONS

- | | |
|-----------------------|---------------------|
| Economic Event ID | Grant / Award |
| Source System | Project |
| Source Transaction ID | Bank Purpose |
| Reporting Fund | Evidence References |
| Source Book | Effective Period |
| Department / Function | Verification State |
| Funding Source | Approval State |

NO PUBLIC SCHEMAS, CODE, PROMPTS, THRESHOLDS OR PRIVATE CONFIGURATION

CONTROL MODEL

AI interprets context. Controls decide. People authorize.

A consistent boundary separates language work, accounting computation and municipal responsibility.

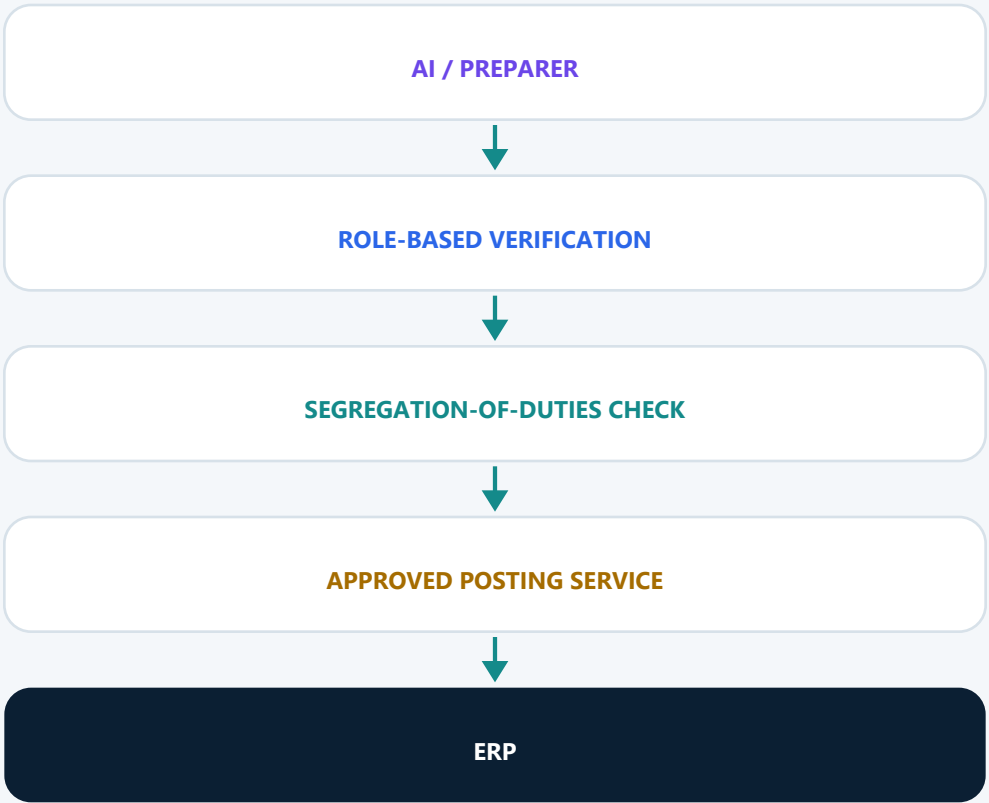
AI-ASSISTED	DETERMINISTIC CONTROLS	HUMAN AUTHORITY
Interpret documents and context Semantic retrieval Propose classifications Draft explanations Assemble review packages	Accounting math Reconciliation + duplicate detection Restriction + budget tests Interfund matching Payroll allocation tests Grant expenditure accumulation Approval validation Posting idempotency	Verify facts Verify authority Exercise accounting judgment Approve actions Accept management responsibility

The model may prepare work. It cannot approve its own work, exercise municipal authority, or hold unrestricted ERP write credentials.

APPROVAL + SEGREGATION OF DUTIES

Approval authority follows role and responsibility - not system access alone.

Transaction type, governing authority and required accounting judgment determine the authorized municipal role.



No self-approval

Preparer and AI cannot approve their own work.

Role required

System access alone does not create authority.

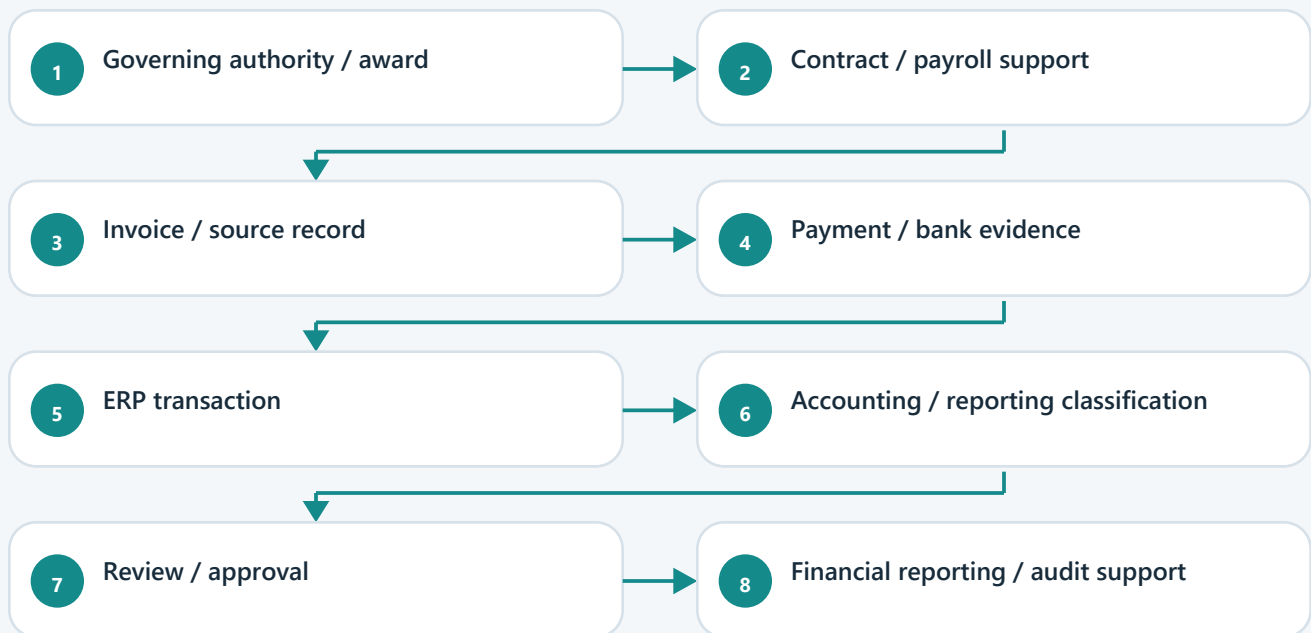
Judgment routes

Complex conclusions go to qualified accounting review.

ECONOMIC EVENT + EVIDENCE LINEAGE

Audit preparation is built into the transaction lifecycle

An Economic Event ID keeps authority, evidence, accounting treatment, payment and reporting connected.



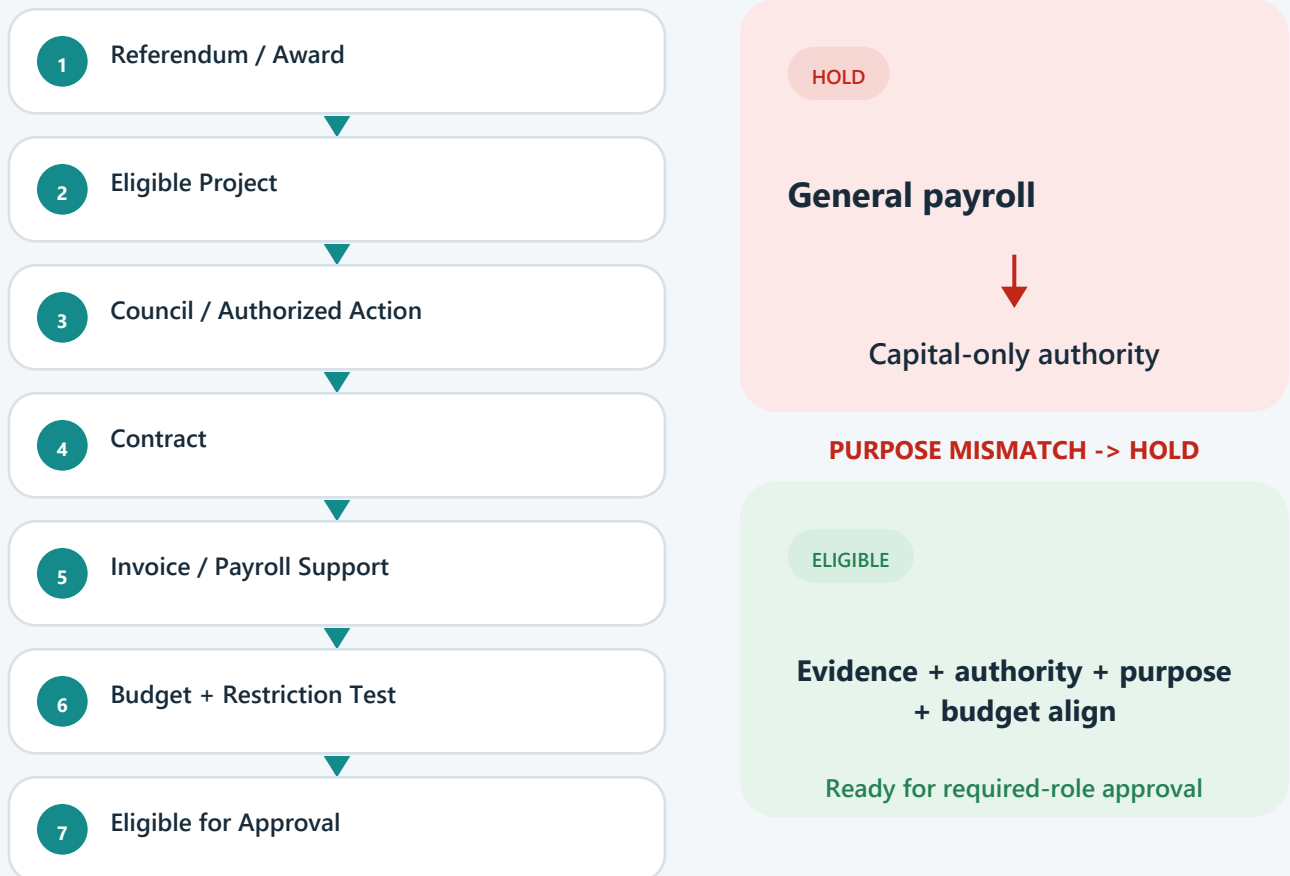
Transaction-to-evidence lineage is built during accounting, reducing year-end reconstruction for auditors.

Management-side outcome: documents, authorization, funding source, restrictions, accounting treatment and payment evidence remain tied to the same economic event.

RESTRICTED FUNDS

Eligibility follows authority - not cash, labels or convenience

The full chain must support purpose, period, project and budget before an item can move to approval.

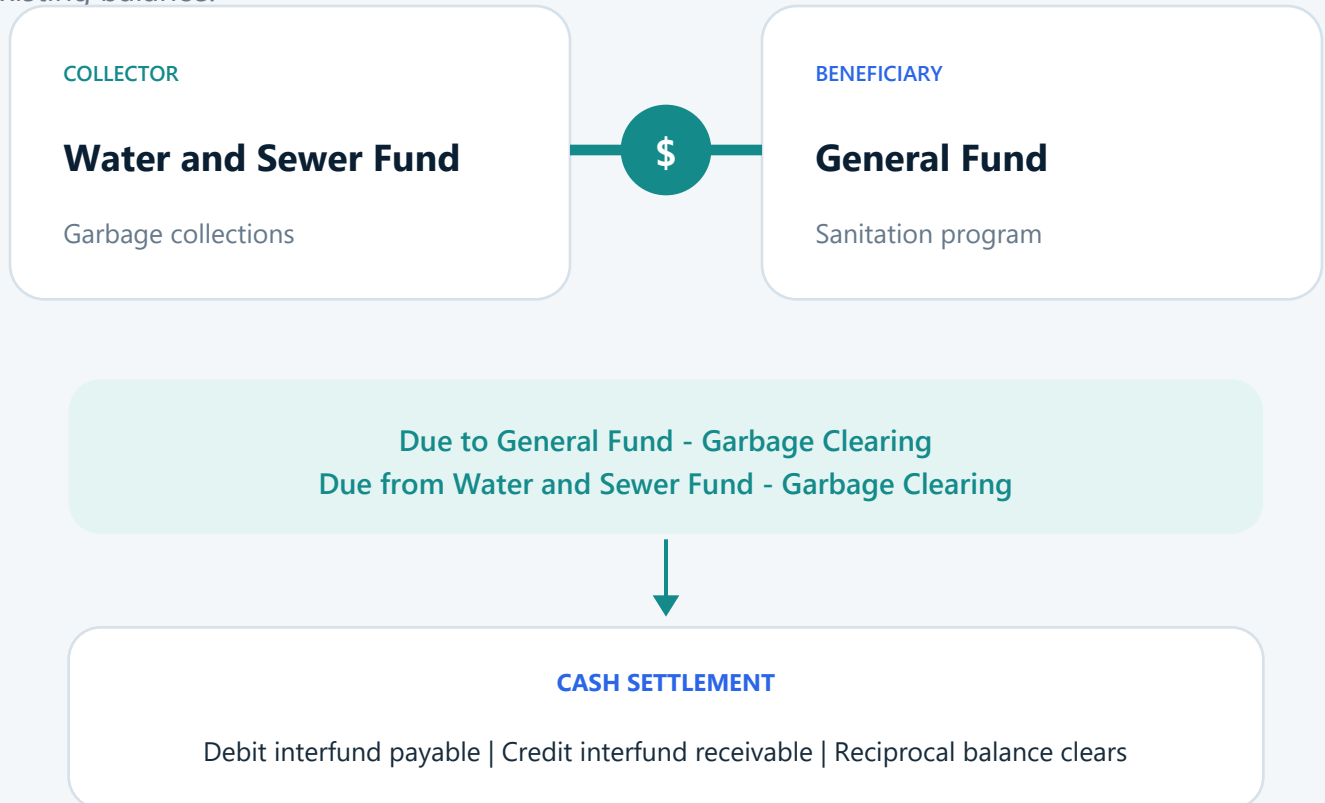


A transfer memo, bank movement or account label cannot override referendum, grant, contract or project authority.

INTERFUND ACCOUNTING

Reciprocal balances preserve the substance of interfund activity

Generic public example: one fund collects on behalf of another, then settlement clears the existing balance.



Settlement of an existing reciprocal balance is not automatically a new transfer or expenditure.

GRANTS + FEDERAL AWARDS + SEFA

Award terms drive recognition, expenditure accumulation and SEFA support



VALIDATION + CONCLUSION

Designed to work with existing municipal software - no replacement ERP required.

Functionally tested with QuickBooks Desktop 2022 in a controlled test environment.

Existing systems

remain the systems of record

Accounting expertise

guides judgment and policy

Controlled AI

interprets and prepares

Deterministic controls

calculate and test

Human authority

verifies and approves

Audit-ready evidence

supports close, reporting and SEFA

The Municipal Finance Control Layer was conceived and developed by Elaina Lockhart as an AI-assisted control architecture for small-government finance.

Synthetic functional prototype. Not an audit, assurance report, legal opinion, peer review, security certification, or production-ready accounting system. Independent governmental CPA design review planned.